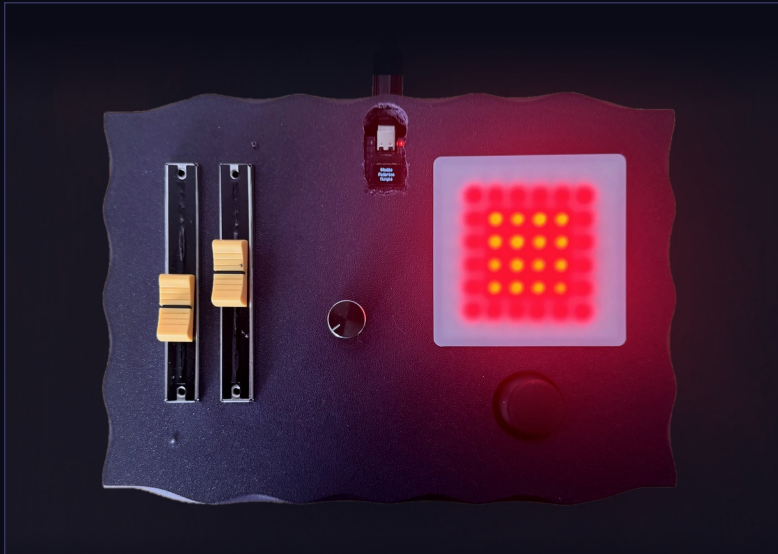


AV SPATIALIZER

ESP32-C3 audiovisual MIDI controller, five controls, 8x8 LED matrix

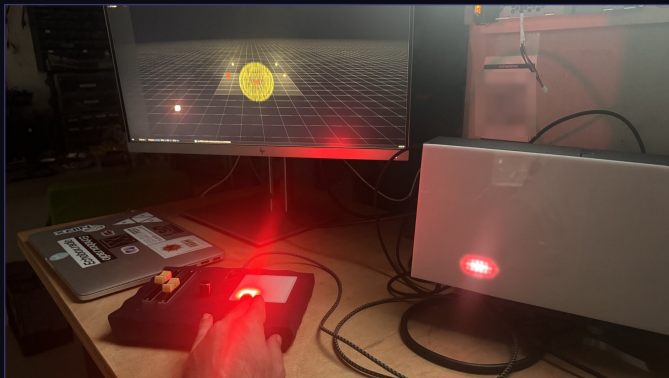
BETA



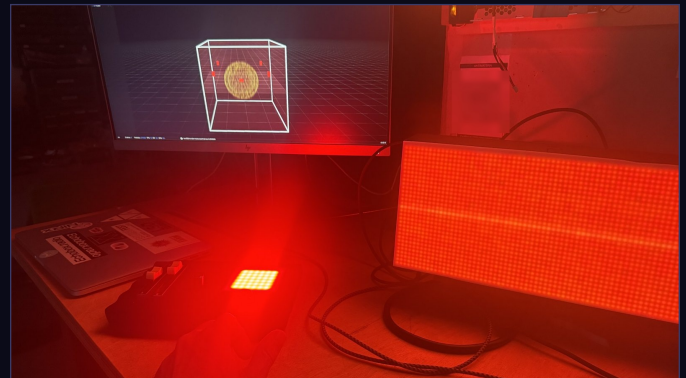
Top view: sliders, potentiometer, 8x8 LED matrix under a frosted diffuser, joystick, OLED display and USB-C.

BORN AS A SPATIALIZER

The controller was first built to move and scale a sound source in 4DSOUND. The joystick places the source in the room, a slider sets its size, and the LED matrix mirrors where the source sits and how large it is.



Moving and scaling a sound source in 4DSOUND, own studio: the source made smaller and moved to the bottom left.

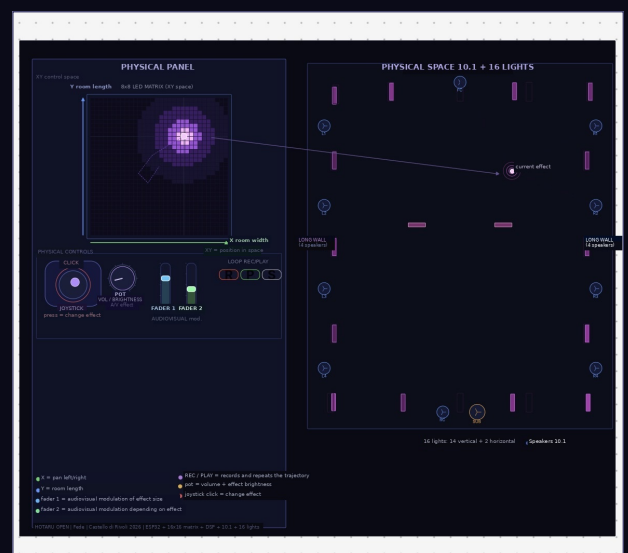


The same source enlarged and left in the centre.

HOTARU OPEN, CASTELLO DI RIVOLI, MAY 2026

It was then configured and performed for Hotaru Open, a residency and public event with John Bringwolves at Castello di Rivoli within the ADRenaline programme, where 15 students from Turin institutions performed spatialised audiovisual compositions with custom-built instruments and interfaces.

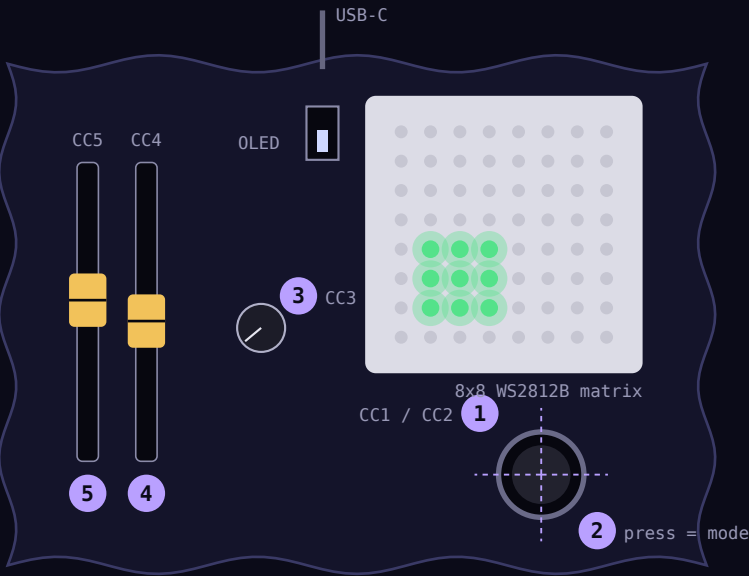
There it drove a 10.1 speaker system and 16 lights (14 vertical, 2 horizontal). The joystick positioned an audiovisual effect in the room, heard and seen in the same place. The click changed effect, the potentiometer set volume and effect brightness together, and the two sliders modulated the effect audiovisually. Joystick trajectories could be recorded and looped in the patch.



Control scheme built and performed for Hotaru Open.

FIVE CONTROLS

MIDI map, onboard behaviour, the Hotaru Open mapping and general-purpose use



HARDWARE AND OUTPUT

ESP32-C3 driving an 8x8 WS2812B LED matrix. Control data goes out as BLE MIDI or over serial, depending on the host. BLE MIDI sends the five parameters as standard Control Change messages. The serial stream uses a tagged format readable by external parsers:

```
sx-...-sy-...-x-...-y-...-p-...  
-s1-...-s2-...-m-...
```

ONBOARD OLED

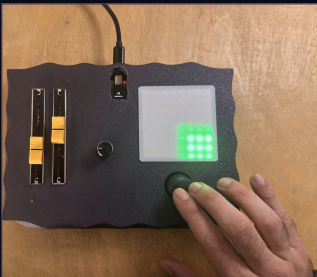
Shows the joystick centre calibration offsets, the X/Y grid position, pot and slider values and the active colour mode, so the controller can be checked without a computer.

EIGHT COLOUR MODES, EIGHT MIDI CHANNELS

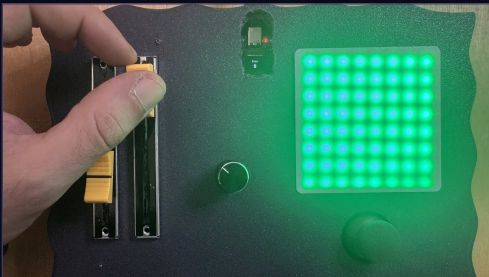


A joystick press steps to the next mode. Each mode sends on its own MIDI channel, so switching mode switches channel and all five CCs follow: one controller addresses eight separate layers in the host.

CONTROL	ON THE PANEL	HOTARU OPEN	GENERAL PURPOSE
1 Joystick X / Y CC 1 / CC 2, 0-127	Horizontal and vertical position of the lit block on the matrix.	X = pan left / right, Y = room length: position of the effect in the room.	Any pair of parameters as an XY pad: source position, filter cutoff and resonance, position of a light or visual element.
2 Joystick press 8 modes = ch 1-8	Steps through the eight colour modes.	Change effect.	Layer, scene or instrument select: all CCs move to the new channel.
3 Potentiometer CC 3, 0-127	LED brightness.	Volume and effect brightness together.	Master level coupling loudness and light intensity, or any continuous parameter.
4 Slider 1 CC 4, 0-127	Block size, from 1x1 to 8x8.	Audiovisual modulation of effect size.	Size or spread of a source, send amount, scale or blur of a visual.
5 Slider 2 CC 5, 0-127	Strobe frequency.	Audiovisual modulation depending on the effect.	Rate parameters: LFO speed, tremolo, strobe or flicker of fixtures.



Joystick moving the block.



Block at full 8x8 size.



Side profile of the enclosure.